Spacetime Algebra as a Theorem: Deriving Cl(3,1) from the Structure of a Dynamical Unit

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Self-contained manuscript; requires no external framework beyond standard linear algebra and Clifford algebra conventions. Numerical verifications referenced in §7 are at
https://github.com/hmolinab/papers/tree/main/weld_clifford/code

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Abstract

We derive the real Clifford algebra $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ from three structural axioms about any operative dynamical unit (ODU). A1 (SAIR): the unit is described by four intrinsic attributes — a scalar S and three vectors $\mathrm{A}, \mathrm{I}, \mathrm{R}$ in $\mathbb{R}^d$.
A2 (geometric product): structure is governed by the geometric product of those attributes, whose grade-2 part $\mathrm{I} \wedge \mathrm{R}$ is the Field bivector. A3 (continuous evolution): the ODU evolves smoothly in time and space at finite propagation speed. From these three axioms — without postulating a spacetime metric or background geometry — we derive: (i) the closure condition $\binom{d}{2} = d$ forces $d = 3$ uniquely (Hodge self-duality of bivectors in $\mathbb{R}^3$, confirmed by Hurwitz); (ii) smooth evolution (A3) requires a fourth temporal direction independent of the spatial attributes; (iii) the principal symbol of the resulting second-order PDE is the Minkowski form $\eta = \mathrm{diag}(-1, +1, +1, +1)$, whose real Clifford algebra is $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$.
Three closing propositions establish that orthonormality is gauge-redundant (P1), that γ_0 is the algebraic generator conjugate to ∂_τ in the Dirac factorization of $\Box$ (P2, without conflating algebra elements with differential operators), and that the Frobenius norm is the unique Clifford inner product forced by A2 (P3). The result promotes the Clifford algebra of spacetime from a geometric postulate to a structural theorem. Classical mechanics and free electrodynamics appear as limits.

Keywords: Clifford algebra, geometric algebra, Hurwitz theorem, spacetime signature, dynamical systems, matrix normal form, Frobenius metric.

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1. Introduction

The Clifford algebra $\mathrm{Cl}_{3,1}$ — equivalently, the spacetime algebra (STA) of Hestenes (1966) — is the standard algebraic scaffolding for special relativity and Dirac theory. Its standard motivation is geometric: one postulates a Minkowski spacetime with signature $(3, 1)$ and then constructs the associated Clifford algebra. The question we address is different: is the Lorentzian signature a theorem, rather than a postulate, if one asks what algebraic structure a self-describing dynamical unit must have?

We show that the answer is yes, under three minimal axioms: A1 (attribute structure), A2 (geometric product), and A3 (smooth evolution at finite speed). The derivation does not require spacetime as an input; the signature emerges from the principal symbol of the equation of motion dictated by A3.

This paper is part of a larger program — the Gamma Space Framework (GSF) — whose central object is a real 4×4 configuration matrix $\Gamma \in M_4(\mathbb{R})$. The present paper establishes the algebraic foundation: that Γ is an element of $\mathrm{Cl}_{3,1}$, not by postulate but by necessity. The companion paper (Molina 2024a) establishes the dynamical result: that the determinant of Γ is the source of the cubic term in the soft-mode reduction of the matrix gradient flow.

Terminology note. Throughout this paper we use the term operative dynamical unit (ODU) as a self-contained technical term requiring no prior knowledge of the GSF. In the GSF literature (Molina 2025), the same object is called a Unit of Coherence (UoC); the two terms are synonymous.

Relation to the geometric algebra literature. The spacetime algebra program (Hestenes 1966, 1986; Doran and Lasenby 2003) is the closest antecedent. That program takes Minkowski spacetime as given and develops physics in terms of $\mathrm{Cl}_{1,3}$ (one time, three space — Hestenes' convention). The signature choice is not merely a convention: $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$ (quaternionic), whereas $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ (real). These are non-isomorphic as real algebras. The present derivation forces $\mathrm{Cl}_{3,1}$ — not $\mathrm{Cl}_{1,3}$ — because we require Γ to be a real matrix (dissipation and gradient flows are real processes); this distinguishes the two conventions at the algebraic level. The derivation does not compete with the spacetime algebra program; it identifies which real algebra is forced by the structure of any evolving dynamical unit, and explains why that program works.

Plan. §2 states the three axioms. §3 derives the four lemmas. §4 states and proves the main theorem. §5 establishes the three closing propositions. §6 illustrates with two physical limits (Newton and Maxwell). §7 gives numerical verification of key steps. §8 discusses scope, related work, and open problems.

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2. Axioms

We consider an operative dynamical unit (ODU): an entity that (i) exists as a coherent whole distinct from its environment, (ii) acts upon its environment, (iii) has an intrinsic drive, and (iv) is embedded in a relational context. No further phenomenological content is assumed.

Axiom A1 (SAIR attribute structure). Any ODU is completely described at the structural level by four intrinsic attributes:
- $S \in \mathbb{R}$ (Singularity, grade 0 — identity / self-measure)
- $\mathrm{A}, \mathrm{I}, \mathrm{R} \in \mathbb{R}^d$ (Agency, Impulse, Relation — grade-1 vectors)

where d is to be determined. The mapping from observable properties of any coherent entity to the four structural slots $\{S, \mathrm{A}, \mathrm{I}, \mathrm{R}\}$ is structurally unique (no two inequivalent assignments produce structurally identical predictions for the same entity).

Remark 2.1. A1 is the foundational axiom of the framework; it is not derived from simpler premises within this paper. Its justification is the structural argument that $\{S, \mathrm{A}, \mathrm{I}, \mathrm{R}\}$ are the grades of a geometric algebra of minimal dimension consistent with A2 — a circularity resolved by the mutual consistency of A1 and A2, not by an independent proof of A1. The role of A1 in this structure is analogous to that of natural selection in Darwinian theory: a minimal posit that generates the rest.

Axiom A2 (geometric product dynamics). The dynamics of an ODU is governed by the geometric product of its attributes. In a geometric algebra $G(d)$ over $\mathbb{R}^d$, the geometric product of two grade-1 elements u, v decomposes canonically:

$$uv = u \cdot v + u \wedge v$$

into a symmetric (grade-0) scalar part and an antisymmetric (grade-2) bivector part. Applied to the attributes: the Force sector Γ_s is the symmetric Gram coupling of the scalar identity S with the grade-1 attributes $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$, encoding the metric structure of the unit. The Field $\mathcal{F} = \mathrm{I} \wedge \mathrm{R} \in \Lambda^2(\mathbb{R}^d)$ (grade-2, antisymmetric) is the reactive sector. "Force" names the symmetric sector of Γ; S is grade-0 and $S\mathrm{A}$ is a grade-1 vector, not a scalar — the symmetric structure enters through the Gram matrix, not through a grade-0 product. The Force/Field split is algebraically forced by A2, not a separate postulate.

Axiom A3 (continuous evolution). The ODU evolves smoothly in time τ and space x, with a finite propagation speed $c > 0$. Treating $\Gamma(\tau, \mathrm{x})$ as a field and expanding to second order in both τ and x, consistent with A1 and A2, the generic equation of motion is

$$\ddot{\Gamma} + \gamma\dot{\Gamma} - c^2\nabla_{\mathrm{x}}^2\Gamma + \nabla_\Gamma P(\Gamma) = N(\Gamma),$$

where $\gamma \geq 0$ is a constitutive damping parameter, $\nabla_{\mathrm{x}}^2 = \partial_{x_1}^2 + \partial_{x_2}^2 + \partial_{x_3}^2$

is the spatial Laplacian (propagation at speed c), and $\nabla_\Gamma P$ is the gradient of the structural potential in configuration space. This is the lowest-order equation coupling inertia ($\ddot{\Gamma}$), spatial propagation ($-c^2\nabla_x^2\Gamma$), dissipation ($\gamma\dot{\Gamma}$), and restoring forces ($\nabla_\Gamma P$); no additional postulate about dynamics is made beyond smoothness and finite speed.

Remark 2.2. The genuine content of A2 is the claim that structure is the geometric product. A3 adds the claim that evolution is smooth and second-order: position and velocity are independent degrees of freedom, so a first-order equation would conflate them. The symmetric/antisymmetric split of the geometric product is a theorem of geometric algebra, not an additional hypothesis.

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3. Four Lemmas

Lemma 1 (Closure — dimension is forced to $d = 3$)

Lemma 1. Under A1 and A2, the dimension of the vector attribute space is $d = 3$.

Proof (closure argument). A2 says the Field is the grade-2 part of the geometric product: $\mathcal{F} = \mathrm{I} \wedge \mathrm{R} \in \Lambda^2(\mathbb{R}^d)$, a bivector of dimension $\binom{d}{2} = \frac{d(d-1)}{2}$. For the ODU to be closed — for $\mathcal{F}$ to couple back to the grade-1 attribute A without introducing objects of higher rank than those in A1 — the Field space and the attribute space must be isomorphic as vector spaces:

$$\binom{d}{2} = d \implies \tfrac{d(d-1)}{2} = d \implies d = 3.$$

The unique non-trivial solution is $d = 3$. This isomorphism is the Hodge duality $\star : \Lambda^2(\mathbb{R}^3) \xrightarrow{\sim} \mathbb{R}^3$, which maps $\mathcal{F} = \mathrm{I} \wedge \mathrm{R}$ to the familiar vector cross product $\mathrm{I} \times \mathrm{R} \in \mathbb{R}^3$.

Confirmation by Hurwitz. Once $d = 3$ is established by closure, the Eckmann theorem (Eckmann 1943; Adams 1960) confirms that a non-degenerate vector cross product on $\mathbb{R}^3$ exists — it is the imaginary part of the quaternion product ($\mathbb{H}$, dimension $d + 1 = 4$). This is a consistency check, not the source of the derivation: Hurwitz verifies that $d = 3$ works, but closure is what forces it.

The octonionic branch ($d = 7$) is structurally distinct. A cross product on $\mathbb{R}^7$ exists (Eckmann 1943) as the imaginary part of the octonion product, but it does not arise from the closure condition $\binom{d}{2} = d$ (since $\binom{7}{2} = 21 \neq 7$). It is not the Hodge dual of a bivector; it is a genuinely different algebraic structure. This branch is the algebraic home of the internal sector ($\mathrm{Der}(\mathbb{O}) = G_2$, three generations) and is treated separately in §8.2. The remainder of this paper works $d = 3$. □

Corollary 1.1. “Why exactly three vector attributes” is not a free parametric choice — it is the answer to “what dimension allows the Field to couple back to the Agents without rank escalation.”

Lemma 2 (Algebra closure)

Lemma 2. In $d = 3$, the three grade-1 attributes $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ generate the full geometric algebra $G(3)$ of dimension $8 = 2^3$.

Proof. Three linearly independent vectors in $\mathbb{R}^3$ generate $G(3)$ by definition: the basis elements are $\{1, e_1, e_2, e_3, e_1e_2, e_2e_3, e_3e_1, e_1e_2e_3\}$ (grade 0 through 3). S occupies grade 0; $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ occupy grade 1; the bivectors (grade 2) and the pseudoscalar (grade 3) are generated by their products. No fifth grade-1 generator is available in $G(3)$: the grade-1 subspace has dimension 3. □

Lemma 3 (Time as a fourth direction)

Lemma 3. Under A3, the smooth evolution of the ODU requires a temporal direction ∂_τ that is independent of the three spatial attribute directions of A1. Together they span a 4-dimensional vector space V^4.

Proof. By A3, the configuration $\Gamma(\tau, \mathrm{x})$ is smooth in time τ and space x. Smoothness implies the existence of partial derivatives ∂_τ and ∇ (the spatial gradient along the attribute directions). Lemma 2 establishes that the spatial attribute space is $\mathbb{R}^3$, spanned by $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. The temporal direction ∂_τ is independent of these three generators for three reasons: (i) it cannot be expressed as a spatial combination of $\mathrm{A}, \mathrm{I}, \mathrm{R}$, since it differentiates the temporal argument, not the attribute frame; (ii) S is grade-0 (scalar), not a grade-1 direction; (iii) promoting one spatial attribute to the temporal role would break the closure condition $\binom{d}{2} = d$ that forces $d = 3$ in Lemma 1. Therefore ∂_τ is a genuinely fourth independent direction.

Together $\{\mathrm{A}, \mathrm{I}, \mathrm{R}, \partial_\tau\}$ span a 4-dimensional vector space V^4. The Clifford algebra $\mathrm{Cl}(V^4, q)$ has dimension $2^4 = 16$; its smallest faithful real matrix representation is 4×4 (Bott periodicity, once the signature q of V^4 is fixed by Lemma 4). The ODU has a time axis because it evolves smoothly — not because spacetime is postulated. □

Lemma 4 (Lorentzian signature from the wave operator)

Lemma 4. The Lorentzian signature $(3, 1)$ is forced by the principal symbol of the EOM (A3), not postulated.

Proof. By A3, the EOM explicitly contains the spatial Laplacian $-c^2\nabla_{\mathrm{x}}^2\Gamma$. The principal part — the highest-derivative terms, which govern propagation — is therefore the wave operator $\Box\Gamma = \ddot{\Gamma} - c^2\nabla_{\mathrm{x}}^2\Gamma$. Isotropy of the Laplacian across the three spatial directions follows directly from P1: since all physical invariants of Γ are $SO(3)$-invariant under rotations of $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$, no spatial direction is preferred, and the spatial part of the principal symbol must be $c^2|\mathrm{k}|^2$ (scalar in k, not a directionally biased tensor). This rules out any anisotropic operator. (A parabolic operator such as the heat equation $\partial_\tau\Gamma = c^2\nabla_{\mathrm{x}}^2\Gamma$ is first-order in time and does not treat ∂_τ as an independent generator on equal footing with the spatial ones; it is excluded.) Taking the Fourier symbol ($\partial_\tau \to i\omega, \nabla \to i\mathrm{k}$):

$$\sigma(\Box) = -\omega^2 + c^2|\mathrm{k}|^2 = \eta^{\mu\nu}p_\mu p_\nu, \quad \eta = \mathrm{diag}(-1, +1, +1, +1)$$

This is the Minkowski quadratic form with signature $(3, 1)$. The real Clifford algebra of this form (Atiyah, Bott and Shapiro 1964; Lounesto 2001) is $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$.

Uniqueness of the real representation. The Bott periodicity classification gives $\mathrm{Cl}_{3,1}$ as a real matrix algebra. Requiring Γ to be a real matrix (the gradient flow $\dot{\Gamma} = -\nabla P$ is real; dissipation is a real process) pins the signature to $(3, 1)$ and excludes $(1, 3)$ (which gives $\mathrm{Cl}_{1,3} \cong M_2(\mathbb{H})$, quaternionic), $(4, 0)$ (which gives $\mathrm{Cl}_{4,0} \cong M_2(\mathbb{H})$, also quaternionic), and all other real signatures that fail to give $M_4(\mathbb{R})$. The only signature that yields a real 4×4 matrix algebra with three spatial and one temporal generator is $(3, 1)$. □

Remark 3.1. The step “$d = 3$ forces real 4×4” is the logical bridge: asking for three spatial generators in a real matrix algebra forces exactly $M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}$. The Lorentzian signature is not a choice — it is what remains after demanding reality and three spatial dimensions.

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4. Main Theorem

Theorem (Γ is forced). Given axioms A1 and A2, the configuration of any operative dynamical unit is necessarily

$$\boxed{\Gamma = \Gamma_s \oplus \Gamma_a \;\in\; M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}}$$

where $\Gamma_s = \frac{1}{2}(\Gamma + \Gamma^\top)$ (symmetric, 10 independent entries — the Force sector) and $\Gamma_a = \frac{1}{2}(\Gamma - \Gamma^\top)$ (antisymmetric, 6 independent entries — the Field sector), with:
- $\Gamma_s = \mathrm{Gram}(\mathrm{A}, \mathrm{I}, \mathrm{R}; S)$ — the symmetric metric coupling structure of the attributes (Force sector; S is the grade-0 identity, not a grade-1 generator)
- Γ_a = magnetic part ($\mathrm{I} \wedge \mathrm{R}$, spatial bivector, from SAIR) $\oplus$ electric part ($\partial_\tau \wedge \nabla$, spacetime bivector, coupling structure↔evolution)

Proof. Lemma 1 forces $d = 3$ by closure. Lemma 2 establishes $G(3)$. Lemma 3 adds the temporal generator ∂_τ, extending $G(3)$ to a 4-generator algebra. Lemma 4 identifies the resulting algebra as $\mathrm{Cl}_{3,1} \cong M_4(\mathbb{R})$ via the wave operator symbol.

Uniqueness of $\Gamma = \Gamma_s \oplus \Gamma_a$. A positive-definite symmetric matrix (metric) carries only Force ($\Gamma_s \succ 0$, no antisymmetric part). A symplectic form carries only Field (Γ_a, no symmetric part). The object $\Gamma = \Gamma_s \oplus \Gamma_a$ is the unique minimal object that carries both Force and Field simultaneously — the unique decomposition of a real 4×4 matrix into symmetric and antisymmetric parts. □

Remark 4.1. The theorem does not claim that $\mathrm{Cl}_{3,1}$ is the algebra of spacetime as a physical background. It claims that any operative dynamical unit that satisfies A1 and A2 has a configuration object that is an element of $\mathrm{Cl}_{3,1}$. Physical spacetime is not an input — it is a limit (the space-time branch of the framework; see §6.2).

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5. Three Closing Propositions

Three technical choices appear in the proof — the orthonormality of the attribute frame, the identification of the temporal generator, and the metric on $M_4(\mathbb{R})$. Each is forced, not free.

Proposition P1 (Orthonormality is gauge-redundant)

Proposition P1. The physical invariants of Γ — its determinant, trace, eigenvalues, and singular values — are independent of the choice of orthonormal frame for $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$. Orthonormality is a representational gauge, not a physical postulate.

Proof. Let $M \in SO(3)$ be the Gram-Schmidt rotation mapping any linearly independent triple $\{\mathrm{A}, \mathrm{I}, \mathrm{R}\}$ to an orthonormal frame $\{\hat{\mathrm{A}}, \hat{\mathrm{I}}, \hat{\mathrm{R}}\}$ spanning the same subspace. Under this rotation, $\Gamma \mapsto M\Gamma M^\top$. The spectrum of Γ (equivalently, $\det\Gamma$, $\mathrm{tr}\,\Gamma$, and all eigenvalues) is invariant under conjugation by any $M \in SO(3)$. The orthonormal frame fixes the explicit matrix entries of Γ_s (diagonal in that frame) but not its spectrum. □

Proposition P2 (The temporal generator is the evolution operator)

Proposition P2. The temporal Clifford generator $\gamma_0 \in \mathrm{Cl}_{3,1}$ satisfying $\gamma_0^2 = -1$ is the unique grade-1 generator assigned to the temporal direction such that the linear differential operator $\mathcal{D} = \gamma^\mu \partial_\mu$ factorizes the wave operator: $\mathcal{D}^2 = \Box$.

Proof. The symbol of the wave operator $\Box = \partial_\tau^2 - c^2\nabla^2$ is $\eta^{\mu\nu}p_\mu p_\nu$ with $\eta = \mathrm{diag}(-1, +1, +1, +1)$. This defines the anticommutation relations $\{\gamma_\mu, \gamma_\nu\} = 2\eta_{\mu\nu}$, with $\gamma_0^2 = -1$ forced by $\eta_{00} = -1$. The assignment of γ_0 to the temporal direction ∂_τ is the unique one that makes $(\gamma^\mu\partial_\mu)^2 = \Box$ (the Dirac factorization). Here γ_0 is an algebra element and ∂_τ is a differential operator; P2 does not claim they are the same object — it claims γ_0 is the algebraic generator conjugate to ∂_τ in the factorization of $\Box$. Verified numerically: a real 4×4

representation satisfying $\{\gamma_\mu, \gamma_\nu\}/2 = \eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$
exists with $\gamma_0^2 = -I$, $\gamma_i^2 = +I$ (residual $< 10^{-14}$; see code/verify_cl31.py). □

Note. P2 does not conflate categories: γ_0 is an element of $\mathrm{Cl}_{3,1}$; ∂_τ is a differential operator acting on functions $\Gamma(\tau, \mathrm{x})$. What P2 establishes is a canonical pairing between the two, mediated by the Dirac factorization of the wave operator given by A3.

Proposition P3 (Frobenius is the canonical Clifford metric)

Proposition P3. The metric on the space of configurations $\Gamma \in M_4(\mathbb{R}) = \mathrm{Cl}_{3,1}$ is the Frobenius norm $\|\Gamma\|^2 = \mathrm{Tr}(\Gamma^\top \Gamma)$, already determined by A2.

Proof. The geometric product of A2 defines the canonical inner product on any Clifford algebra:

$$\langle A, B \rangle_{\mathrm{Cl}} := \langle A\tilde{B} \rangle_0$$

the grade-0 (scalar) component of the geometric product of A with the Clifford reverse $\tilde{B}$. In the real 4×4 representation of $\mathrm{Cl}_{3,1}$, the Clifford reverse corresponds to matrix transposition ($\tilde{B} = B^\top$), giving $\langle A, B \rangle_{\mathrm{Cl}} = \frac{1}{4}\mathrm{Tr}(A^\top B)$, which is the Frobenius inner product up to the normalization $\frac{1}{4}$.

Uniqueness. $\mathrm{Spin}(3, 1)$ acts on $\mathrm{Cl}_{3,1}$ by adjoint conjugation $g \cdot \Gamma = g\Gamma g^{-1}$. Any physical metric must be invariant under this action. The grade decomposition $\Lambda^0 \oplus \Lambda^1 \oplus \Lambda^2 \oplus \Lambda^3 \oplus \Lambda^4$ of $\mathrm{Cl}_{3,1}$ consists of pairwise non-isomorphic irreducible representations of $\mathrm{Spin}(3, 1)$. By Schur's lemma, any $\mathrm{Spin}(3, 1)$-invariant bilinear form is proportional to $\mathrm{Tr}(A^\top B)$ on each grade-block, with a possibly different constant on each grade. The submultiplicativity constraint — $\|\Gamma\Gamma'\| \le \|\Gamma\|\|\Gamma'\|$ for all $\Gamma, \Gamma' \in \mathrm{Cl}_{3,1}$, required for $P(\Gamma) = \|\Gamma\|^2$ to be compatible with the algebra product — forces equal scaling across all grades (a different scale per grade would violate submultiplicativity for mixed-grade products). The result is Frobenius, uniquely. □

Extension to $\mathrm{Cl}_{4,1}$. The spacetime configuration in the GSF framework lives in $\mathrm{Cl}_{4,1} \cong M_4(\mathbb{C})$ (Bott periodicity: $p - q = 3$). The same argument applies with the Clifford reverse replaced by the Hermitian conjugate: $\langle A, B \rangle = \frac{1}{4}\mathrm{Tr}(A^\dagger B)$, giving the Hilbert-Schmidt norm $\|\Gamma\|^2 = \mathrm{Tr}(\Gamma^\dagger \Gamma)$, real and non-negative. P3 holds without modification.

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6. Two Physical Limits

The main theorem and propositions establish the algebraic structure. Two physical theories appear as limiting cases, showing the algebra has concrete content.

6.1 Newton's second law (Force sector, det > 0)

In the operationally active sector ($\det \Gamma > 0$, $\Gamma_s \succ 0$), the dominant dynamics is along the soft mode of Γ — the direction of smallest singular value. Project the EOM onto the soft-mode scalar x (dominant singular value of Γ along A):

$$\ddot{x} + \gamma\dot{x} + \partial_x P = F_{\mathrm{ext}}$$

This is Newton's second law for a damped oscillator: mass is the inertia of the soft mode, γ is the damping of the ODU, and P is the potential landscape. Newton's law is not a special case built into the framework — it is what the EOM becomes when projected onto the dominant mode. Status: structural correspondence $\langle \mathrm{CE} \rangle$, not a derivation from first principles.

6.2 Free electrodynamics (Field sector, det = 0 boundary)

At the boundary $\det\Gamma = 0$, the configuration loses rank: one or more singular values of Γ approach zero. When $\gamma \to 0$ (non-dissipative limit) and the configuration lives in the grade-2 (bivector) sector Γ_a of $\mathrm{Cl}_{3,1}$, the EOM reduces to:

$$\Box\Gamma_a = 0 \quad \Rightarrow \quad \Box F_{\mu\nu} = 0$$

which is the free Maxwell equation in the absence of sources ($\partial^\mu F_{\mu\nu} = 0$), together with the Bianchi identity $\partial_{[\mu}F_{\nu\rho]} = 0$ (automatic from $F = \mathrm{d}A$). The bivector $F_{\mu\nu}$ is the Faraday tensor; the identification of Γ_a with the grade-2 sector of $\mathrm{Cl}_{3,1}$ makes the Field I ∧ R correspond to the spatial (magnetic) part of $F_{\mu\nu}$, and the electric part arises from the $\partial_\tau \wedge \nabla$ coupling. Charge conservation $\partial^\mu J_\mu = 0$ follows from the antisymmetry of F without additional assumptions. Status: free Maxwell ⟨TEO⟩[D]; source equation ⟨A⟩ (requires closing the cross-coupling block).

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7. Numerical Verification

The following steps in the derivation are numerically confirmed; scripts are in code/:

Result	Script	Residual
$\{\gamma_\mu, \gamma_\nu\}/2 = \eta_{\mu\nu}$ in real 4×4 rep	verify_cl31.py	$< 10^{-14}$
$\gamma_0^2 = -I$, $\gamma_i^2 = +I$	verify_cl31.py	$< 10^{-14}$
$\langle A, B\rangle_{\mathrm{Cl}} = \mathrm{Tr}(A^\top B)/4$ on grade-1 elements	verify_clifford_metric.py	$< 10^{-14}$
Frobenius submultiplicativity: $\lVert\Gamma\Gamma'\rVert_F \le \lVert\Gamma\rVert_F \lVert\Gamma'\rVert_F$ (0 violations); Pythagorean: $\lVert\Gamma\rVert^2 = \lVert\Gamma_s\rVert^2 + \lVert\Gamma_a\rVert^2$	verify_frobenius.py	0 violations; error $< 10^{-13}$
$\det\Gamma$ as invariant under $SO(3,1)$ conjugation	verify_det_invariance.py	$< 10^{-12}$

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8. Discussion

8.1 What is new

The spacetime algebra program (Hestenes 1966; Doran and Lasenby 2003) takes $\mathrm{Cl}_{3,0}$ or $\mathrm{Cl}_{1,3}$ as the algebra of physical space or spacetime, motivated by the known geometry. The present work reverses this logic: $\mathrm{Cl}_{3,1}$ is derived as the forced algebraic structure of any self-describing dynamical unit, without assuming a spacetime background. The key steps are:
(i) the closure condition $\binom{d}{2} = d$ forces the dimension of the vector attribute space (Hurwitz confirms consistency);
(ii) the wave operator fixes the signature.
Neither step is obvious from the physics-first perspective.

The closest antecedent is the observation (Lounesto 2001, §17) that the Clifford algebra of the symbol of the wave operator is $\mathrm{Cl}_{3,1}$. Here that observation becomes a theorem: it is the only Clifford algebra consistent with A1 and A2.

8.2 The octonionic branch

Lemma 1 identifies a second branch: $d = 7$ (octonions). As argued in Lemma 1, this branch does not admit a canonical temporal extension compatible with the octonionic product structure. It is the algebraic home of the internal structure of the ODU: the derivation algebra $\mathrm{Der}(\mathbb{O}) = G_2$

(with $\mathrm{SU}(3) \subset G_2$), three generations encoded in the exceptional Jordan algebra $h_3(\mathbb{O})$, and color symmetry. The two branches ($d = 3$ with time, $d = 7$ without) are algebraically orthogonal: the temporal generator of the $d = 3$ branch does not act on the $d = 7$ sector. This structural orthogonality may explain why internal quantum numbers appear to be independent of spacetime dynamics, but the connection remains conjectural and is open for future work.

8.3 Honest scope

This paper establishes the algebraic structure. It does not:
- Derive the specific metric of physical spacetime (GR is a limit; the derivation requires additional steps developed in Molina 2025, Part V)
- Prove uniqueness of the SAIR attribute structure independent of A1 (that is the content of A1 itself, which we take as foundational rather than derived)
- Close the $d = 7$ (octonionic) branch identification (open problem; partial results in Molina 2025, §Q)

The residues that remain open after P1/P2/P3 are exactly A1, A2, and A3 — the three axioms. Everything else in the derivation is a theorem. The cost of explicitness: three axioms instead of two. The gain: no premiss enters the derivation undeclared.

8.4 Related work

- Hurwitz (1898): normed division algebras in dimensions 1, 2, 4, 8.
- Eckmann (1943): cross products in $\mathbb{R}^n$ exist only for $n = 1, 3, 7$.
- Atiyah, Bott and Shapiro (1964): Clifford modules and Bott periodicity.
- Hestenes (1966, 1986): spacetime algebra as the language of physics.
- Doran and Lasenby (2003): geometric algebra for physicists (Cambridge).
- Lounesto (2001): Clifford algebras and spinors (Cambridge).
- Adams (1960): vector fields on spheres; confirms Hurwitz via K-theory.
- Molina (2024a): the determinant as the source of the cubic term in matrix gradient flows. DOI: 10.5281/zenodo.20752208

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9. Conclusions

From three structural axioms — SAIR attribute structure (A1), geometric product (A2), and continuous smooth evolution (A3) — the real Clifford algebra $\mathrm{Cl}_{3,1}$ emerges as a structural theorem rather than a geometric postulate. The derivation chain is:

$$\underbrace{\text{SAIR}}_{\text{A1}} + \underbrace{\text{geom. product}}_{\text{A2}} \xrightarrow{\binom{d}{2}=d} d = 3 \xrightarrow{G(3)} \{\mathrm{A}, \mathrm{I}, \mathrm{R}\} = \gamma_i \xrightarrow{\text{A3: smooth, finite } c} \partial_\tau \perp \mathbb{R}^3 \xrightarrow{\sigma(\Box)=\eta} (3,1) \xrightarrow{\text{P1/P2/P3}} \Gamma = \Gamma_s \oplus \Gamma_a \in M_4(\mathbb{R})$$

Three propositions close the technical residues: orthonormality is a representational gauge (P1); γ_0 is the unique algebraic generator conjugate to ∂_τ in the Dirac factorization of $\Box$ — not the operator itself (P2); Frobenius is the canonical Clifford metric forced by A2 (P3). The irreducible axioms are A1, A2, and A3. Classical mechanics (§6.1) and free electrodynamics (§6.2) appear as structural limits.

The result supplies a structural foundation for the spacetime algebra program: not “given Minkowski space, use $\mathrm{Cl}_{3,1}$”, but “from the structure of a dynamical unit, $\mathrm{Cl}_{3,1}$ is the forced representation.”

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